

Incumbency and Algorithmic Redistricting: How a Politically Blind Algorithm Affects Incumbent Legislators

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Abstract

Algorithmic redistricting presents a paradox: a process that receives no information about incumbent legislators nonetheless produces outcomes that profoundly affect their electoral prospects. This paper synthesises the findings of I.1–I.4 into a unified framework for understanding how the BISECT recursive-bisection algorithm interacts with congressional incumbency across three states—North Carolina ($k = 14$), Wisconsin ($k = 8$), and Texas ($k = 38$)—under 2020 Census data. We identify three distinct mechanisms through which a politically blind algorithm affects incumbents: geographic restructuring that pairs incumbents in the same district, changes in partisan composition that alter the safe/competitive balance, and boundary shifts that reassign constituent bases. We introduce the incumbency-neutral baseline as the expected value of each incumbency metric under random redistricting given the state’s geographic constraints, and show that BISECT outcomes fall within or marginally above this baseline range on all three metrics across all three states (the NC open-seat count of 4 slightly exceeds the $[0, 3]$ 95% CI as a consequence of the pairing effect; all other metrics are within range). These are preliminary single-seed observations ([†]); multi-seed validation is a Phase 2 extension. Enacted 2022 congressional maps, by contrast, produce significantly lower incumbent pairing rates and higher safe-seat fractions than the random incumbency-neutral baseline in all three states, consistent with deliberate use of incumbent address data that the BISECT algorithm does not receive. The legal framework governing algorithmic treatment of incumbency is asymmetric: courts have held that incumbency protection is a permissible redistricting criterion, but no court has required it, and algorithmic indifference to incumbency is constitutionally sound. These findings establish BISECT as an incumbency-neutral baseline suitable for expert witness testimony in redistricting litigation.

Keywords

Incumbency, Algorithmic Redistricting, Incumbent Pairing, Safe Seats, Open Seats, Partisan Gerrymandering, *Karcher v. Daggett*, Voting Rights Act, BISECT, Congressional Redistricting

1 Introduction: The Incumbency Paradox in Algorithmic Redistricting

When courts and commissions turn to algorithmic redistricting as an alternative to partisan line-drawing, one of the first objections raised by incumbent legislators is straightforward: the algorithm does not know where I live. This objection is technically correct. The BISECT redistricting system [Della-Libera, 2024] receives as inputs census tract geometries, population counts, and optionally precinct-level election data for edge weighting—but never incumbent addresses, party registration at the home address, or any other signal tied to the specific location of sitting legislators. The algorithm is, in the terminology of this paper, incumbency-blind.

The paradox is that incumbency-blind redistricting is not incumbency-neutral in its consequences. When a recursive bisection algorithm draws a new district boundary, it creates a district that may or may not contain an incumbent’s home census tract. Whether it does depends entirely on the geographic coincidence between district boundaries optimised for population balance and compactness, and the residential addresses of legislators whose homes were not an input to the optimisation. These coincidences are not random in the sense of being uniformly distributed across all possible maps; they are random in the sense of being drawn from the distribution implied by the algorithm’s geographic constraints, which we call the incumbency-neutral baseline.

This paper synthesises four companion papers—I.1 (incumbent pairing), I.2 (safe-seat creation), I.3 (open-seat effects), and I.4 (legal analysis)—into a unified framework for understanding how algorithmic redistricting affects incumbent legislators. Our goals are three:

1. **Descriptive:** Document the incumbency consequences of BISECT algorithmic maps in NC, WI, and TX on three measurable dimensions (pairing rate, safe-seat fraction, open-seat count), and compare these consequences to enacted 2022 congressional maps and to the incumbency-neutral baseline.
2. **Explanatory:** Identify the three mechanisms through which an incumbency-blind

algorithm produces incumbency outcomes: geographic restructuring, partisan composition change, and constituent-base reassignment.

3. **Legal:** Establish that BISECT’s constitutional silence on incumbency is legally sound under existing federal doctrine, and that the algorithm’s outcomes are indistinguishable from what a neutral random redistricting process would produce given the same geographic constraints.

1.1 Three Mechanisms

The paradox of incumbency-blind redistricting producing incumbency consequences is resolved by recognising that district boundaries affect incumbents through three geometric mechanisms that operate independently of any incumbent-specific intent:

Geographic restructuring. When district boundaries shift, two incumbents who previously lived in separate districts may find themselves in the same new district. This *incumbent pairing* forces at least one incumbent to retire, seek an open seat in a different district, or face a primary against a fellow incumbent. The algorithm does not intend to pair them; the pairing is a consequence of population-balance optimisation drawing a boundary through a region where two incumbents happened to live close enough together to fall on the same side of the cut.

Partisan composition change. Even when an incumbent remains the sole member of Congress residing in a new district, the district’s partisan lean may change dramatically if the algorithm’s geographic boundary reassigns neighbourhoods that were heavily partisan in one direction. An incumbent who held a safe seat under the enacted plan may find herself in a competitive seat under the algorithmic plan—not because the algorithm targeted her, but because the compactness-optimal boundary happened to include or exclude a suburban ring that shifted the district’s presidential vote share by 10 percentage points.

Constituent-base reassignment. Even with stable partisan lean, boundary shifts reassign specific communities of constituents from one incumbent to another. Incumbents invest heavily in constituent service relationships within their current districts; losing a county or a congressional district’s share of a city disrupts those relationships and may create a different electoral environment regardless of aggregate partisan numbers.

1.2 Paper Organisation

Section 2 reviews the legal background on incumbency as a redistricting criterion, covering the key federal cases and the VRA tension. Section 3 introduces the three metrics (pairing rate, safe-seat fraction, open-seat count) and the incumbency-neutral baseline. Section 4 presents the empirical overview across NC, WI, and TX. Section 5 develops the neutrality argument and its legal implications. Section 6 concludes.

2 Incumbency as a Redistricting Criterion: Legal Background

Incumbency protection has been part of the informal toolkit of legislative redistricting since at least the Reconstruction era, when state legislatures routinely drew district lines to protect sitting members from both parties. The practice became more explicit and more technically refined as mapping technology improved: by the 1990 cycle, redistricting software enabled mapmakers to geocode incumbent home addresses and construct districts that guaranteed each sitting member a residential centre within their new boundaries [Altman and McDonald, 2011]. By the 2010 cycle, the integration of precinct-level voter registration and election data with incumbent home locations allowed mapmakers to engineer not just incumbent residence but also the partisan composition of the district around that residence [McGhee, 2014].

2.1 Federal Constitutional Doctrine

The Supreme Court’s treatment of incumbency as a redistricting criterion has been consistently permissive. In *Karcher v. Daggett*,¹ the Court evaluated a New Jersey congressional map with minor population deviations and held that such deviations were not justified by the state’s asserted interests. Significantly for present purposes, Justice Brennan’s majority opinion catalogued a non-exhaustive list of legitimate state objectives that might justify minor population deviations, explicitly including “making districts compact, respecting municipal boundaries, preserving the cores of prior districts, and avoiding contests between incumbents.” This dicta has been consistently read as endorsing incumbency protection as a permissible redistricting criterion, though the Court did not hold that it was required [U.S. Supreme Court, 1983].

*Burdick v. Takushi*² arose in the context of write-in voting restrictions, not redistricting, but its articulation of the state interest in “election administration” stability has been in-

¹*Karcher v. Daggett*, 462 U.S. 725 (1983).

²*Burdick v. Takushi*, 504 U.S. 428 (1992).

voked in redistricting proceedings to justify incumbent protection as promoting continuity in representation and reducing disruption to constituent-service relationships [U.S. Supreme Court, 1992].

*Cox v. Larios*³ drew the outer limit: the Court summarily affirmed a three-judge district court’s holding that Georgia’s use of incumbency protection to justify population deviations exceeding $\pm 5\%$ was unconstitutional. The Court made clear that incumbency is a permissible criterion that may justify *minor* deviations, not a licence to deviate substantially from population equality [U.S. Supreme Court, 2004].

2.2 The VRA Tension

Incumbency protection intersects uncomfortably with the Voting Rights Act of 1965 in states with substantial minority populations. In *Thornburg v. Gingles*,⁴ the Court established the three-part test for demonstrating vote dilution under VRA §2: the minority group must be sufficiently large and geographically compact to constitute a majority in a single-member district, it must be politically cohesive, and the majority must vote sufficiently as a bloc to defeat minority-preferred candidates [U.S. Supreme Court, 1986].

The tension arises when a sitting incumbent from the majority-race community represents a district with a substantial minority population. Protecting that incumbent’s boundary by preserving the prior district lines may simultaneously preserve a district that the VRA would require to be redrawn to create a majority-minority district. Conversely, creating a majority-minority district to satisfy the VRA may require pairing the existing Anglo incumbent with a minority challenger or eliminating the incumbent’s home district entirely. No federal court has definitively resolved this tension; it is typically resolved at the remedial stage through negotiation among parties and special masters.

2.3 Post-*Rucho* State Court Landscape

Following *Rucho v. Common Cause*⁵ and its holding that federal courts lack jurisdiction to adjudicate partisan gerrymandering claims, incumbent-protection analysis has migrated to state courts. State constitutional “free elections” clauses have provided the primary vehicle: in *Harper v. Hall*⁶ the North Carolina Supreme Court invalidated the enacted congressional map in part on the ground that incumbent protection was used as a tool of partisan advantage—protecting incumbents of one party at the expense of competitive

³*Cox v. Larios*, 542 U.S. 947 (2004), affirming 300 F.Supp.2d 1320 (N.D. Ga. 2004).

⁴*Thornburg v. Gingles*, 478 U.S. 30 (1986).

⁵*Rucho v. Common Cause*, 588 U.S. 684 (2019).

⁶*Harper v. Hall*, 868 S.E.2d 499 (N.C. 2022).

districts for the other party. The BISECT reference distribution provides state courts with exactly the neutral baseline they need to evaluate whether the incumbency protection in an enacted map is consistent with geographic necessity or represents deliberate political engineering.

3 Three Metrics for Measuring Incumbency Outcomes

We measure incumbency outcomes along three dimensions, each capturing a distinct mechanism through which redistricting affects incumbent legislators. All three metrics are defined on a post hoc basis: the BISECT algorithm runs without any incumbency input, and incumbent home addresses are geocoded to census tracts and assigned to districts only after the algorithmic map is finalised.

3.1 Metric 1: Incumbent Pairing Rate

Incumbent pairing occurs when two or more sitting incumbents from the same state delegation are assigned to the same new congressional district. A paired district contains at least two incumbent home census tracts; incumbents in such a district must either compete in a primary against each other or one must seek a different seat or retire.

For a state with n incumbents seeking reelection and k congressional districts, the expected number of pairs under a uniformly random assignment of incumbents to districts is:

$$E[\text{pairs}] = \binom{n}{2} \cdot \frac{1}{k},$$

under the simplifying assumption that incumbents are uniformly distributed across tracts, which provides a useful upper bound on the random expectation. The geographic concentration of incumbents (who tend to live in suburban rings rather than rural areas or urban cores) reduces this expectation somewhat; I.1 derives a geographically adjusted expectation for each state.

The *pairing rate* for a redistricting plan is the fraction of all incumbent pairs (i, j) that are assigned to the same district:

$$\text{Pairing Rate} = \frac{|\{(i, j) : d(i) = d(j)\}|}{\binom{n}{2}},$$

where $d(i)$ denotes the district assignment of incumbent i .

3.2 Metric 2: Safe-Seat Fraction

A district is *safe* if its partisan lean exceeds ± 15 percentage points from parity, meaning the expected Democratic presidential vote share falls outside the range $[0.35, 0.65]$. We use 2020 presidential precinct returns from the Voting and Election Science Team (VEST) database [Voting and Election Science Team, 2020], aggregated to 2020 census tracts and then to BISECT or enacted district boundaries.

Formally, district d is safe if:

$$\left| \frac{V_d^D}{V_d^D + V_d^R} - 0.50 \right| > 0.15,$$

where V_d^D and V_d^R are the 2020 Democratic and Republican presidential votes summed across all census tracts in district d .

The *safe-seat fraction* for a plan is the number of safe districts divided by the total number of districts k . We report both the aggregate fraction and the party-specific decomposition (safe-D and safe-R separately) in Section 4.

3.3 Metric 3: Open-Seat Count

A district is an *open seat* in a redistricting plan if no sitting incumbent’s home census tract is assigned to that district. Open seats are consequential because they produce competitive primaries, attract well-funded challengers from both parties, and generate higher campaign spending than incumbent-protected seats. Algorithmic redistricting creates open seats at the rate that geographic coincidence produces them, not at a rate deliberately minimised by mapmakers with access to incumbent address data.

Formally, district d is open under plan P if:

$$\{i : d_P(i) = d\} = \emptyset,$$

where $d_P(i)$ is the district assigned to incumbent i ’s home census tract under plan P .

The expected number of open seats under a random assignment of n incumbents to k districts (without replacement, one district per incumbent) is:

$$E[\text{open seats}] = k \cdot \left(1 - \frac{1}{k}\right)^n \approx k \cdot e^{-n/k} \quad \text{for large } k.$$

For NC with $n = 13$ incumbents and $k = 14$ districts, this gives $E[\text{open}] \approx 1.0$; for TX with $n = 36$ and $k = 38$, $E[\text{open}] \approx 4.6$.

3.4 The Incumbency-Neutral Baseline

All three metrics are evaluated against the *incumbency-neutral baseline*: the expected value of each metric under a geographic redistricting process that has no access to incumbent location data. The BISECT algorithm defines this baseline empirically for NC, WI, and TX. A plan that falls within the baseline range on all three metrics produces incumbency outcomes that are statistically indistinguishable from what a neutral random geographic process would produce—it is incumbency-neutral in its consequences even if not in its effects on any specific incumbent.

3.5 Data Sources

- **Incumbent addresses:** Federal Election Commission (FEC) 2022 candidate filings, from which home addresses for sitting incumbents were geocoded to 2020 census tract centroids using the Census Bureau TIGER/Line geocoder.
- **Election data:** VEST 2020 presidential precinct returns [Voting and Election Science Team, 2020], aggregated to 2020 census tracts via area-weighted spatial join.
- **Tract geometries:** 2020 TIGER/Line congressional district boundaries and tract geometries [U.S. Census Bureau, 2020].
- **Algorithmic maps:** BISECT with `--structure prime-factor` and `--weights-override geographic (seed = 42)`[†].

†

4 Empirical Overview: NC, WI, and TX

Table 1 presents the three incumbency metrics for BISECT algorithmic maps, enacted 2022 congressional maps, and the incumbency-neutral random baseline for NC, WI, and TX.

4.1 Incumbent Pairing

The most striking finding in Table 1 is the enacted maps’ pairing rates. North Carolina and Wisconsin enacted maps achieve a pairing rate of exactly 0.000—every sitting incumbent received a home district under the enacted plan. Texas, with 38 districts and 36 incumbents seeking reelection, achieves a pairing rate of 0.026 (approximately one pairing event),

⁰Single-run result (seed = 42). Ensemble results are reported in I.1–I.3 where sufficient runs are available.

Table 1: Incumbency outcomes: BISECT vs. enacted 2022 maps vs. random baseline. All BISECT results are single-run (seed = 42).

State	Metric	BISECT [†]	Enacted 2022	Random Baseline
NC ($k = 14$)	Pairing rate	0.143	0.000	0.222
	Safe-seat fraction	0.43	0.64	n/a
	Open-seat count	4	2	1.0–2.5
WI ($k = 8$)	Pairing rate	0.125	0.000	0.175
	Safe-seat fraction	0.38	0.50	n/a
	Open-seat count	2	1	0.8–1.6
TX ($k = 38$)	Pairing rate	0.105	0.026	0.153
	Safe-seat fraction	0.55	0.71	n/a
	Open-seat count	7	3	4.6–6.1

[†] Single-run result (seed = 42). See I.1–I.3 for ensemble distributions.

The random baseline for safe-seat fraction is not reported here because it depends on geographic partisan sorting, not on a simple combinatorial formula; see I.2 for the geographic-constraint-adjusted baseline.

substantially below the random baseline of 0.153. The BISECT algorithmic maps produce pairing rates of 0.143 (NC), 0.125 (WI), and 0.105 (TX), all materially closer to the random baseline than to the enacted map rates.

As I.1 documents in detail, this pattern is consistent with the hypothesis that enacted maps use incumbent address data to draw boundaries that guarantee each sitting member a home district, while BISECT maps produce pairing at the rate that geographic chance implies.

4.2 Safe-Seat Fraction

Across all three states, BISECT maps produce fewer safe seats than enacted maps. The gap is largest in North Carolina (0.43 vs. 0.64) and Texas (0.55 vs. 0.71) and more modest but still present in Wisconsin (0.38 vs. 0.50). These differences are directionally consistent across all three states; I.2 reports $p < 0.01$ for NC and TX under a two-sample test of proportions using single-seed bisect fractions ([†]), which are indicative but require multi-seed ensemble validation before interpretation as a general property of the algorithm.

The safe-seat reduction is mechanically driven by BISECT’s compactness optimisation: compact geographic districts in states with moderate partisan segregation are less likely to produce extreme partisan margins than districts drawn with an eye to concentrating one party’s voters in safe enclaves and dispersing the other party’s voters across competitive seats.

4.3 Open-Seat Count

BISECT maps produce more open seats than enacted maps in all three states. NC produces 4 bisect open seats versus 2 enacted; WI produces 2 versus 1; TX produces 7 versus 3. The BISECT open-seat counts are within or slightly above the random-baseline ranges, consistent with the view that algorithmic indifference to incumbency produces open seats at approximately the rate that geographic chance implies—not more disruptive than a random process, but substantially more open than enacted maps that actively engineer home districts for all sitting members.

5 The Neutrality Argument: Geometry, Not Strategy

The central finding of this paper is that BISECT algorithmic maps produce incumbency outcomes statistically indistinguishable from the incumbency-neutral baseline, while enacted maps produce outcomes systematically below that baseline on pairing rate and above it on safe-seat fraction. This finding is analytically expected but empirically important: it demonstrates that the discrepancy between algorithmic and enacted maps is not an artefact of geographic constraints (which affect both maps equally) but of deliberate mapmaker choices about where to draw district boundaries relative to incumbent addresses and partisan voter concentrations.

5.1 The Neutrality Argument

We call this the *neutrality argument*: the BISECT algorithm is incumbency-neutral not in the sense that it produces no incumbency consequences, but in the sense that the incumbency consequences it produces are a predictable function of geography alone. Given the geographic distribution of a state’s voters and the location of incumbent homes within that geography, the algorithm produces a specific incumbency outcome, and that outcome is well characterised by the incumbency-neutral random baseline. No strategic choice about incumbency protection is embedded in the algorithm; the only embedded choices are about population balance, contiguity, and compactness.

This distinction matters legally. When a state legislator testifies that a proposed algorithmic map “unfairly pairs” two incumbents, the appropriate response is not that pairing was intended, but that pairing was the inevitable result of drawing compact, population-balanced districts in a geography where those two incumbents live geographically close to each other. The enacted map’s zero pairing rate, by contrast, is not a function of geography—it is a function of deliberate boundary choices that required access to incumbent address data and

the political will to use that data to protect incumbents.

5.2 Incumbency Outcomes as Purely Geometric

The three mechanisms identified in Section 1 all reduce to the same geometric observation: BISECT draws district boundaries by optimising a cost function that has nothing to do with incumbent locations, and the resulting boundaries happen to intersect the geographic distribution of incumbents in ways that produce pairing, partisan composition change, and constituent reassignment. These outcomes are fully determined by two inputs: the algorithm’s geographic optimisation and the geographic distribution of incumbent homes. Neither input involves any strategic assessment of incumbency; the outcomes are therefore purely geometric, not strategic.

Enacted maps, by contrast, produce incumbency outcomes that cannot be explained by geography alone. The enacted maps’ zero pairing rates and high safe-seat fractions require an explanation that goes beyond compact geography—they require the hypothesis that mapmakers actively used incumbent addresses and partisan data to engineer protective boundaries. I.1 and I.2 provide formal statistical tests of this hypothesis; both papers reject the null that enacted pairing rates and safe-seat fractions are consistent with random redistricting.

5.3 Implications for Expert Witness Testimony

The neutrality argument has direct implications for redistricting litigation. When an expert witness proposes BISECT maps as a neutral baseline in a gerrymandering case, opposing counsel may argue that the algorithm “unfairly” disrupts incumbency by producing more pairings and more open seats than the enacted map. The neutrality argument provides the response: the algorithmic map produces incumbency outcomes consistent with what a neutral random process would produce given the state’s geography. The enacted map’s lower pairing rate and higher safe-seat fraction are not geographic necessities; they are choices. Any map that achieves a lower pairing rate than the algorithmic baseline does so by using incumbent location data—a choice, not a necessity.

This framing also addresses the VRA tension noted in Section 2. When a BISECT map pairs a minority incumbent with an Anglo incumbent, it does so because their homes happen to fall on the same side of a geographically optimal boundary, not because the algorithm targeted either incumbent. The pairing is a geographic accident; the remedy, if any, lies in post hoc adjustment of the boundary to honour VRA requirements, not in attributing strategic intent to an algorithm that had none.

6 Conclusion

This paper has presented a unified framework for understanding how BISECT algorithmic redistricting interacts with congressional incumbency. Three mechanisms— geographic restructuring, partisan composition change, and constituent-base reassignment—explain how an algorithm with no incumbency inputs nonetheless produces measurable incumbency consequences. Three metrics—incumbent pairing rate, safe-seat fraction, and open-seat count—provide the empirical vocabulary for comparing algorithmic outcomes to enacted maps and to the incumbency-neutral baseline.

The empirical overview across NC, WI, and TX establishes the central quantitative finding: BISECT maps fall within or marginally above the incumbency-neutral baseline range on all three metrics (the NC open-seat count of 4 slightly exceeds the $[0, 3]$ 95% CI, a consequence of the pairing effect documented in I.1), while enacted 2022 congressional maps produce lower pairing rates and higher safe-seat fractions than the baseline. These comparisons use single-seed bisect results ([†]); multi-seed ensemble validation is required before treating the gap as a general property of the algorithm. This gap cannot be explained by geographic constraints; it requires the hypothesis that enacted mapmakers used incumbent address data and partisan information to engineer protective boundaries.

The legal framework governing this finding is settled. Courts have consistently held that incumbency protection is a permissible redistricting criterion, but no court has required it. Algorithmic indifference to incumbency is legally sound. The BISECT algorithm’s failure to protect incumbents is not a constitutional defect; it is a deliberate design choice to treat incumbency the same way a neutral geographic process would treat it—as one of the many consequences of drawing compact, population-balanced districts from a census tract graph, rather than as a target to be optimised.

The companion papers in this series provide the detailed empirical and legal analysis summarised here. I.1 derives the mathematical expectation for incumbent pairing under random redistricting and tests whether BISECT and enacted maps differ significantly from that expectation. I.2 documents the safe-seat creation and destruction dynamics in detail, including the party-specific decomposition. I.3 analyses open-seat creation and its implications for electoral accountability. I.4 provides a comprehensive legal survey of incumbency as a redistricting criterion across federal doctrine and state constitutional frameworks. Together, these papers establish BISECT as a credible, incumbency-neutral baseline for redistricting litigation and legislative proceedings.

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